Step by Step Guide
Competitive Pokémon Analysis: A Graph-Based Approach to Exploring Usage and Strategies

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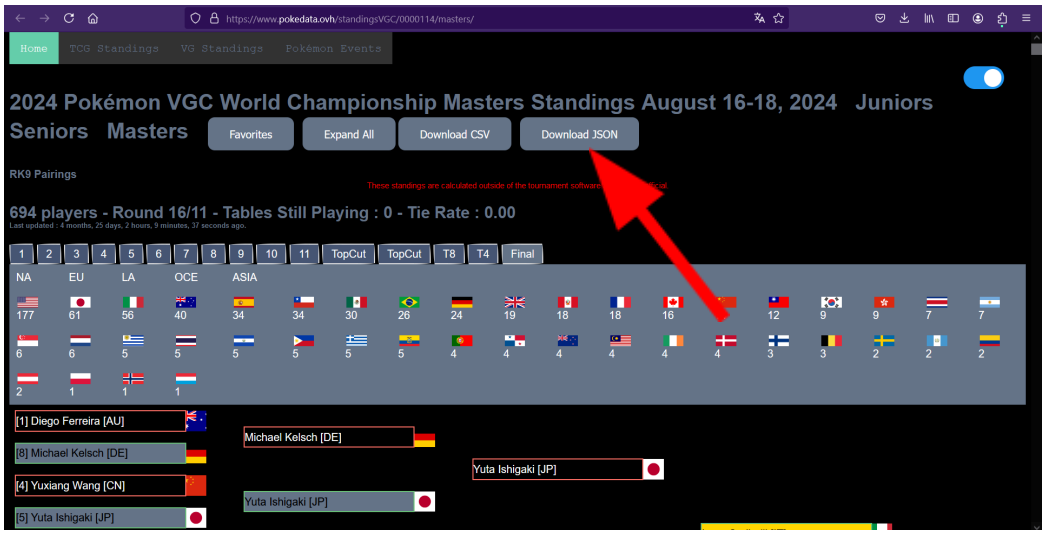
Dataset used and Acquisition

Using the [pokeapi.ipynb]() notebook, retrieve information about Pokémon, including their names, type(s), and statistics, and generate a CSV file containing 1,302 rows (records) and 9 columns (attributes).

From the PokeDATA website ([link]()), download theJSON file containing information about players, their records (wins, losses, ties), and the teams they played with. Each team is composed of six Pokémon, characterized by their Teratype, nature, and statistics.
The JSON file contains 693 JSON objects.



Data cleansing

Using the [fix_names.ipynb]() notebook, resolve discrepancies between the Pokémon names retrieved from the PokeAPI and those obtained from the downloaded PokeDATA file.

Then, with [jsonToCSV.ipynb]() notebook, remove irrelevant data from the PokeDATA file and convert the remaining data into CSV format with the following structure: rank, player_name, pokemon_name, teratypes, and nature. This process produces a dataset consisting of 1,164 rows and 5 columns.

Data quality

For the completeness analysis, use the completeness.ipynb notebook to evaluate the completeness of the pokemon_data and player_data datasets. The script performs an analysis of column and table completeness for both datasets.

Data Model and Storage

For the Neo4j database, a Docker version was used, specifically the one provided on the course's eLearning page by the professor.
To create the Neo4j container, run the following command in the directory containing the docker-compose file:
docker-compose up -d neo4j

Data Integration and Enrichment

Using the nodegen3.ipynb notebook, generate a .cypher file containing all the Cypher instructions needed to create the nodes and relationships required to populate the database.
Then, with the neo4jconnect.ipynb notebook, connect to the database and upload the .cypher file.

Queries

Using the queries.ipynb notebook, connect to the database and execute the queries.
Alternatively, you can run each query directly through the Neo4j visual interface by connecting to localhost:7474.